Diabetes Prediction Project Proposal

Project Planning and Management

Program	DEPI
Planning milestone due	16 October 2026

Project Overview

This proposal defines a four-week plan for a student machine-learning project using the diabetes dataset supplied by the team. The project will explore the health indicators in the dataset and build a model that classifies the target field, diabetes, as 0 or 1. The result is an educational prototype and is not intended for medical diagnosis.

Objective

Prepare the supplied data, identify useful patterns, train and compare suitable classification models, and report their performance with clear evaluation metrics.

Project Description

The team will inspect the dataset, document its fields, check data quality, and analyze the relationship between the available health indicators and the diabetes outcome. After preprocessing, the team will train supervised classification models, compare their results, and present the findings in a notebook, report, and final presentation.

Dataset and Prediction Target

The team-provided dataset includes gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level, and diabetes. The prediction target is diabetes, recorded as a binary label (0 or 1).

Tools and Technologies

Python in Jupyter Notebook or Google Colab; pandas and NumPy for data handling; Matplotlib and Seaborn for visualizations; and scikit-learn for preprocessing, model training, and evaluation. The team can replace these choices if the course requires specific tools.

Team Members and Responsibilities

Team member	Primary responsibility
Haneen Alaa Gamal	Project coordination, scope definition, and progress tracking
Shahenaz Amer Gamal	Data understanding, cleaning, and exploratory data analysis
Mohamed Said Sadan	Classification model development and tuning
Mohab Elsayed Godah	Visualizations, results interpretation, and report preparation
Marium Mostafa Fathy	Literature review, model testing, and presentation integration

Milestones from the Project Schedule

Milestone	Planned output	DEPI deadline
Project planning and management	Project proposal and team plan	16 Oct 2026
Literature review and requirements gathering	Short review and documented project requirements	16 Oct 2026
System analysis and design	Data workflow and model design	6 Nov 2026
Implementation	Prepared notebook and model comparison	30 Nov 2026
Final presentation, testing, and reports	Tested project, final report, and presentation	4 Dec 2026

Four Week Work Plan

The plan below follows the four-week structure in the provided template and aligns its milestone dates with the DEPI schedule shown in the screenshot.

Week 1 | Planning, Literature Review, and Requirements | Due 16 Oct 2026

Tasks

- Confirm the project scope, target, and expected outputs.
- Review relevant references on diabetes classification.
- Inspect dataset fields and target values.
- Agree on tools, team roles, and schedule.

Deliverables

- Proposal with project scope, tools, team roles, and schedule.
- Literature notes, requirements, and dataset field summary.

Week 2 | System Analysis and Design | Due 6 Nov 2026

Tasks

- Check missing values, duplicates, ranges, category values, and target balance.
- Decide on data cleaning, categorical encoding, and train/test split approach.
- Draw the project workflow from data loading through evaluation.
- Select candidate classification models and define evaluation metrics.

Deliverables

- Data quality summary and exploratory charts.
- Preprocessing and model workflow design.
- Written analysis and design notes.

Week 3 | Implementation | Due 30 Nov 2026

Tasks

- Clean and preprocess the dataset, then create a reproducible training and testing split.
- Train baseline classification models such as Logistic Regression, Decision Tree, and Random Forest.
- Compare models using accuracy, precision, recall, F1-score, and a confusion matrix.
- Prepare readable charts and document the main findings.

Deliverables

- Runnable Jupyter/Colab notebook with documented steps.
- Model comparison table and evaluation visualizations.
- Initial results summary.

Week 4 | Testing, Final Report, and Presentation | Due 4 Dec 2026

Tasks

- Rerun the notebook from start to finish and check outputs, plots, and reproducibility.
- Review the model results and explain limitations, including that the project is not a medical diagnostic tool.
- Complete the written report and organize the presentation.
- Proofread the final materials and rehearse the presentation.

Deliverables

- Final tested notebook and project files.
- Final report and presentation slides.
- Clear explanation of the model, evaluation results, and project limitations.